

## EXERCISE SHEET 3

### GROUP AND RING THEORY

#### MAIN QUESTIONS - TO BE DONE BEFORE THE SEMINAR

- 3.1.** Find the number of necklaces with six beads and two colours. [Assume that you cannot flip the necklaces, for instance if the beads are different on the reverse side.]
- 3.2.** Let  $G$  be a group of order 168, and assume that  $G$  has a normal subgroup  $N$  of prime order. Show that  $G$  has a normal Sylow  $p$ -subgroup, for some prime divisor  $p$  of  $|G|$ .
- 3.3.** Let  $G$  be a group of order 216. Show that  $G$  is not simple.

#### EXTRA QUESTIONS

- 3.4** Let  $G$  be a non-abelian group of order  $p^3$ , where  $p$  is a prime. Show that  $Z(G) = G'$ . What is the order of  $Z(G)$ ?
- 3.5.** We say that a group  $G$  is *solvable* if there are subgroups  $1 = G_0 < G_1 < \cdots < G_n = G$  such that  $G_{i-1}$  is normal in  $G_i$ , and  $G_i/G_{i-1}$  is an abelian group, for  $i \in \{1, 2, \dots, n\}$ .  
Let  $G$  be a non-abelian group of order  $pq$ , where  $p$  and  $q$  are distinct primes. Show that  $G$  is solvable.